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PAST PAPERS (FALL 2022)

(i) Find the polar form of the complex number

$z = -2 + 2i$

Also find z^5

For Polar form find Magnitude (r) or Argument (θ)

For Magnitude (r)

$r = |z|$

$r = \sqrt{(-2)^2 + (2i)^2}$

$r = \sqrt{4 + 4}$

$r = \sqrt{8}$

$r = 2\sqrt{2}$

For Argument (θ)

$\theta = \frac{2i}{-2} = \theta = -1$

value is $\left(\frac{3\pi}{4}\right)$ $\theta = -1$ $\boxed{-1 \text{ lies in 2nd Quardant}}$

For Find z^5 we'll use De Moivre's

theorm's $\quad z^n = r^n(\cos(n\theta) + i\sin(n\theta))$

$$z^5 = (2\sqrt{[illegible]})^5 (\cos 5\theta + i \sin 5\theta)$$

$$z^5 = (2\sqrt{[illegible]})^5 (\cos 5\theta + i \sin 5\theta)$$

$$z^5 = (2\sqrt{[illegible]})^5 \left\{\cos(-5)(3)\frac{3\pi}{4} + i \sin 5\left(\frac{3\pi}{4}\right)\right\}$$

$$z^5 = (2\sqrt{[illegible]})^5 \left\{\cos\left(\frac{15\pi}{4}\right) + i \sin\left(\frac{15\pi}{4}\right)\right\}$$

$$z^5 = 32\sqrt{2}$$

~~$z^5 = (2\sqrt{[illegible]})^5 \; -\cos(53^\circ) - i\sin(53^\circ)$~~

~~11.3137 4177341.9 $-0.6018 - 0.7986i$~~

~~$z^5 = 4117340.5i$~~

$$z^5 = 32\sqrt{2}\left(\cos\frac{7\pi}{4} + i \sin\frac{7\pi}{4}\right)$$

ii) **Evaluate**

$$\int 5x^2(x^5 + 12)\,dx$$

$$= \int (5x^7 + 60x^2)\,dx$$

$$= 5\int x^7\,dx + 60\int x^2\,dx$$

$$= 5\frac{x^8}{8} + 60\frac{x^3}{3}\,dx$$

$$= \frac{5}{8}x^8 + 20x^3 + c$$

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(iii) Evaluate by Integral By Parts.

$$\int \frac{2x}{(8-x)^3}\,dx$$

$$X\Big(= \int 2x\,(8-x)^{-3}\,dx$$

$$= (8-x)^{-3}\,2\int x\,dx - \int\left\{\frac{d}{dx}(8-x)^{-3}\,2\int x\,dx\right\}dx$$

$$= (8-x)^{-3}\,\cancel{2}\,\frac{x^2}{\cancel{2}} - \int\left\{-3(8-x)^{-3-1}(-1)\,\cancel{2}\,\frac{x^2}{\cancel{2}}\right\}dx$$

$$= \frac{x^2}{(8-x)^3} - \int\left\{3(8-x)^{-4}\,x^2\right\}dx$$

$$= \frac{x^2}{(8-x)^3} - 3\int x^2\,(8-x)^{-4}\,dx$$

$$= \frac{x^2}{(8-x)^3} - 3\int \frac{x^2}{(8-x)^4}\,dx \Big)\,X$$

$$= 2\int \frac{x}{(8-x)^3}\,dx$$

$$= 2\int x\,(8-x)^{-3}\,dx$$

Using By Parts —

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$$= 2\left[x \int (8-x)^{-3} - \int \left\{ \frac{d}{dx}(x) \int (8-x)^{-3} \right\} dx \right]$$

$$= 2\left(x \frac{(8-x)^{-3+1}}{-3+1} - \int 1 \frac{(8-x)^{-3+1}}{-3+1} dx \right)$$

$$= \cancel{2}\left(x \frac{(8-x)^{-2}}{\cancel{-2}} \right) - \cancel{2} \int \frac{(8-x)^{-2}}{\cancel{-2}} dx$$

$$= x(8-x)^{-2} - \frac{(8-x)^{-2+1}}{-2+1} + c$$

$$= \frac{x}{(8-x)^2} - \frac{(8-x)^{-1}}{-1} + c$$

$$= \frac{x}{(8-x)^2} - \frac{1}{8-x} + c$$

(iv) The Marginal Cost is given

$MC = 25 + 30Q - 9Q^2$

$FC = 55$

Find TC, AC or VC

$MC = 25 + 30Q - 9Q_2$

For TC we Integrate the MC

$$\int (MC) = \int 25 + \int 30Q - \int 9Q^2$$

$$TC = 25Q + \frac{30Q^2}{2} - \frac{9Q^3}{3} + c$$

$$TC = 25Q + 15Q^2 - 3Q^3$$

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$$TC = 25Q + 15Q^2 - 3Q^3 + c$$

$$TC + FC$$

$$\boxed{TC = 25Q - 15Q^2 - 3Q^3 + 55}$$

For $AC = \frac{TC}{Q}$

$$AC = \frac{25Q - 15Q^2 - 3Q^3 + 55}{Q}$$

$$AC = \frac{25Q}{Q} - \frac{15Q^2}{Q} - \frac{3Q^3}{Q} + \frac{55}{Q}$$

$$\boxed{AC = 25 - 15Q - 3Q^3 + \frac{55}{Q}}$$

For VC

$$VC = TC - FC$$

$$\boxed{VC = 25Q - 15Q^2 - 3Q^3 - 55}$$

(V) Given demand functions

$P_d = 133 - Q^2$ ——— (i)

or Supply functions

$P_s = (Q+1)^2$ ——— (ii)

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Under Pure Competitions Find Consumer Surplus OR Producer Surplus.

$$P_d = P_s$$

$$133 - Q^2 = (Q+1)^2$$

$$133 - Q^2 = Q^2 + 2Q + 1$$

$$0 = Q^2 + Q^2 + 2Q + 1 - 133$$

$$2Q^2 + 2Q - 132 = 0$$

$$2(Q^2 + Q - 66 = 0$$

$$Q^2 + Q - 66 = \frac{0}{2}$$

$$Q^2 + Q - 66 = 0$$

Using Quardatic formula :-

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{(1)^2 - 4(1)(-66)}}{2(1)}$$

$$= \frac{-1 \pm \sqrt{1 + 264}}{2}$$

$$= \frac{-1 \pm \sqrt{265}}{2}$$

$$= \frac{-1 \pm 16.278}{2}$$

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$$\frac{-1+16.278}{2}, \quad \frac{-1-16.278}{2}$$

$$= \frac{15.278}{2}, \quad \frac{-17.278}{2}$$

$$= 7.635, \quad -8.639$$

So $\boxed{Q_0 = 7.635}$

For P_0 Put in eq (ii)

$$P_0 = (7.3635 + 1)^2$$

$$\boxed{P_0 = 69.948}$$

Now Consumer's Surplus

$$\int_0^{Q_0} f(Q)\, dQ - Q_0 P_0$$

$$= \int_0^{7.635} (133 - Q^2)\, dQ - 7.635(69.948)$$

$$\left[133Q - \frac{Q^3}{3}\right]_0^{7.635} - 534.052$$

$$= \left[133(7.635) - 133(0) - \frac{(7.635)^3}{3} - \frac{0}{3}\right.$$

$$- 534.052$$

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$= 1015.455 - 0 - 148.356 - 5340.52$

$$\boxed{C.S = -4473.921}$$

Now Producer Surplus

$$\int_0^{Q_0} Q_0 P_0 - f(Q)\, dQ$$

$$= \int_0^{7.635} 534.052 - (Q+1)^2\, dQ$$

$$= 534.052 - \int_0^{7.635} (Q^2 + 2Q + 1)\, dQ$$

$$= 534.052 - \left[\frac{Q^3}{3} + \frac{2Q}{2} + Q\right]_0^{7.635}$$

$$= 534.052 - \frac{(7.635)^3 - 0}{3} + 7.635 - 0 + 7.635$$

$$= 534.052 - \frac{148.356}{3} + 15.27$$

$$\boxed{P.S = 400.966}$$

(vi) Solve the differential equations

$$\frac{dy}{dt} - 6y = 18$$

$$\frac{dy}{dt} - 6y = 18$$

$$\frac{dy}{dt} - y = \frac{\cancel{18}\ 3}{\cancel{6}}$$

$$\frac{dy}{dt} - y = 3$$

$$dy - y = 3\,dt$$

First we'll indentity of integrals factor which is (e^{-6t})

(Multiply both side by e^{-6t})

$$e^{-6t}\left(\frac{dy}{dt}\right) - 6e^{-6t}y = 18e^{-6t}$$

Now dividing by equation e^{-6t}

$$\frac{\cancel{e^{-6t}}}{\cancel{e^{-6t}}}\frac{dy}{dt} - \frac{6e^{-6t}}{e^{-6t}}y = \frac{18\cancel{e^{-6t}}}{\cancel{e^{-6t}}}$$

$$= \frac{dy}{dt} - \frac{6e^{-6t}}{e^{-6t}}y = 18$$

$$= \frac{dy}{dt} - 6e^{-6t}y = 18e^{-6t}$$

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$$e^{-6t}\frac{dy}{dt} - 6e^{-6t}y = 18e^{-6t}$$

Integrate both side w.r.t to t

$$\int e^{-6t}\frac{dy}{\cancel{dt}}\cancel{dt} - 6e^{-6t}y = \int 18e^{-6t}$$

$$\frac{\cancel{-6}\,e^{-6t}}{\cancel{-6}}y = \frac{18e^{-6t}}{-6} + c$$

$$e^{-6t}y = \frac{-18}{6}e^{-6t} + c$$

$$e^{-6t}y = \frac{-18}{6}e^{-6t} + c$$

solve for y

$$y = \frac{-3e^{-6t} + c}{e^{-6t}}$$

$$y = \frac{-3\cancel{e^{-6t}}}{\cancel{e^{-6t}}} + \frac{c}{e^{-6t}}$$

$$y = -3 + ce^{6t}$$

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Q#2 Following Questions are long Questions.

(iii) Evaluate the Integral

$$\int x^3 \sin x \, dx$$

using By Parts

$= x^3 \int \sin x \, dx - \int \left\{ \frac{d}{dx}(x^3) \int \sin x \, dx \right\} dx$

$x^3 \; -\cos x - \int (3x^2 \; -\cos x) \, dx$

$-x^3 \cos x + 3 \int (x^2 \cos x) \, dx$

$= -x^3 \cos x + 3 \left[x^2 \int \cos x \, dx - \int \left\{ \frac{d}{dx}(x^2) \int \cos x \right.\right.$

$-x^3 \cos x + 3 \left[x^2 \sin x - \int 2x \sin x \right] dx$

$-x^3 \cos x + 3 \left[x^2 \sin x - 2 \int (x \sin x) \right] dx$

$-x^3 \cos x + 3 \left[x^2 \sin x - 2 \left\{ x \int \sin x - \int \left(\frac{d}{dx}(x) \int \sin x \right) \right. \right.$

$-x^3 \cos x + 3 \left[x^2 \sin x - 2 \left\{ -x \cos x - \int \sin x \right\} \right] dx$

$-x^3 \cos x + 3 \left[x^2 \sin x - 2 \left\{ -x \cos x + \cos x \right\} \right]$

$-x^3 \cos x + 3x^2 \sin x + 6x \cos x - 6 \cos x + C$

$3x \cos x \left(-x^2 + x \sin x \cancel{+2} \cancel{-2} + c \right)$

$= 3x\cos x(-x^2 + x\sin x) + c$

$= 3x\cos x(x\sin x - x^2) + c$

(i) Derive the differential Equations of Solow growth Model for cost and labour Problem in the Term of single variable $\frac{K}{L}$.

DO Your Self

(ii) Solve the second order differential Equations

$$y''_{(t)} - 4y'_{(t)} - 5y_{(t)} = 35$$

$$y_0 = 5 \qquad y'_{(0)} = 6$$

$$a_1 = -4 \qquad a_2 = -5 \qquad b = 35$$

Gernal Equations

$$\therefore \; y_p = \frac{b}{a_2}$$

$$y_t = y_c + y_p$$

$$y_p = \frac{-35}{5} \qquad \boxed{y_p = -7}$$

For y_c use I^{st} Case of (Discrite Root Real case)

$$a_1^2 > 4a_2$$

$$16 > -20$$

So :- $A_1 e^{r_1 t} + A_2 e^{r_2 t}$

— Now we find $r_1 r_2$ —

For r_1, r_2 we use factorai:

$$r^2 - 4r - 5 = 0$$

$$r^2 - 5r + r - 5 = 0$$

$r(r-5)+1(r-5)=0 \Rightarrow (r-5)(r+1)$

$r-5=0\ ,\ r+1=0 \Rightarrow \boxed{r_1=5,\ r_2=-1}$

putt values of r

$Y_c = A_1 e^{5t} + A_2 e^{-1(t)} \Rightarrow \boxed{A_1 e^{5t} + A_2 e^{-t}}$ G.S

Now $y=0$

$Y_0 = A_1 e^0 + A_2 e^0 \Rightarrow A_1 + A_2$

$\because Y_0 = 5$

$\boxed{5 = A_1 + A_2}$ — i)

Now Differciate (Y_c)

$Y'_{(c)} = 5A_1 e^{5t} + (-A_2 e^{-t}) \Rightarrow Y'_{(c)} = 5A_1 e^{5t} - A_2 e^{-t}$

$\because Y'_{(0)} = 6$ if $y=0$

$Y'_{(0)} = 5A_1(1) - A_2(1)$

$\boxed{6 = 5A_1 - A_2}$ — ii)

For eq (i)	Put (A_1) in eq (ii)	Put in eq (ii)
$A_1 + A_2 = 6$	$A_1 + 4 = 6$	$5A_1 - A_2 = 6$
$A_1 = -A_2 + 6$	$A_1 = 6 - 4$	$5(6-A_2) - A_2 = 6$
$\boxed{A_1 = 6 - A_2}$	$\boxed{A_1 = 2}$	$30 - 5A_2 - A_2 = 6$
		$30 - 6A_2 = 6$
		$30 - 6 = 6A_2$
		$24 = 6A_2$
		$A_2 = 24/6$
		$\boxed{A_2 = 4}$

Now Our ~~Definite~~ General Solution is

$Y_t = Y_c + Y_P$

$Y_t = A_1 e^{5t} + A_2 e^{-t} + (-7)$

$\boxed{Y_t = 2e^{5t} + 4e^{-t} - 7}$

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Q#1:- Solve the following:-

(i) Find the Polar form the complax Number $z = 1+2i$ Also z^5.

For Polar form we find Magnitude or argument

Magnitude = r

argument = θ

$r = \sqrt{(1)^2 + (1)^2}$

$r = \sqrt{2}$

For Argument

$\tan\theta = \frac{1}{1} = 1$

So $\theta = 1$ is Ist Quardant value is $\frac{\pi}{4}$

$z = (\sqrt{2})\left[\cos\frac{\pi}{4} + \sin\frac{\pi}{4}\right]$

Now we find z^5
we use De Moivres theorem:

$$z^5 = (\sqrt{2})^5 \left(\cos 5\frac{\pi}{4} + \sin 5\frac{\pi}{4}\right)$$

$$z^5 = 2^{5/2}\left(\cos 5\frac{\pi}{4} + \sin \frac{5\pi}{4}\right)$$

$$z^5 = 4\sqrt{2}\left(\cos 225^\circ + \sin 225^\circ\right)$$

$$z^5 = 4\sqrt{2}\left(-0.7071 - 0.7071\right)$$

$$z^5 = 4\sqrt{2}\left(-1.4142\right)$$

$$z^5 = 2.828\left(-1.4142\right)$$

$$\boxed{z^5 = -3.999}$$

(ii) find the Particular or Gernal Solutions of Differential Equations.

$$y_t + 7y_{t-1} + 6y_{t-2} = 42$$

Gernal Equation :- $Y_t = Y_c + Y_p$

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For $Y_p = \frac{b}{1+a_1+a_2} = \frac{42}{1+7+6} = \boxed{3}$

for Y_c

$a_1^2 > 4a_2 = 49 > 24$

So $Y_c = A_1(r_1)^t + A_2(r_2)^t$

for r_1 or r_2

$r^2 + 7r + 6 = 0 \Rightarrow r^2 + 6r + r + 6$

$r(r+6) + 1(r+6) = 0 \quad (r+1)(r+6) = 0$

$r+1=0 \quad r+6=0 \quad \boxed{r_1 = -1} \quad r_2 = -6$

putt in Y_c

$$\boxed{Y_c = A_1(-1)^t + A_2(-6)^t}$$ G.S

If Y is 0 Then

$Y_0 = A_1(-1)^0 + A_2(-6)^0$

$Y_0 = A_1 + A_2$

So Our Gernal solutions.

$Y_t = Y_c + Y_p$

$$Y_t = A_1(-1)^t + A_2(-6)^t + 3$$

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(iii) Evalute Integral by parts

$$\int \frac{3x}{(x^2-6)^5}\,dx$$

[redacted]

$$= \int \frac{3x}{(x^2-6)^5}\,dx$$

$$= \int \frac{2x+x}{(x^2-6)^5}\,dx$$

$$= \int \frac{2x}{(x^2-6)^5} + \int \frac{x}{(x^2-6)^5}\,dx$$

$$= \int 2x\,(x^2-6)^{-5} + \frac{1}{2}\int \frac{2x}{(x^2-6)^5}\,dx$$

$$= \frac{(x^2-6)^{-5+1}}{-5+1} + \frac{1}{2}\int 2x\,(x^2-6)^{-5}$$

$$= \frac{(x^2-6)^{-4}}{-4} + \frac{1}{2}\,\frac{(x^2-6)^{-4}}{-4} + C$$

$$= \frac{4}{(x^2-6)^4} + \frac{4}{2}\,\frac{1}{(x^2-6)^4} + C$$

$$= \frac{4}{(x^2-6)^4} + \frac{2}{(x^2-6)^4} + C$$

$$= \frac{4 + 2 + \cancel{\ }}{(x^2+6)^4} + C$$

$$= \frac{6}{(x^2+6)^4} + C$$

(iv) Find TC, AC, If
$MC = 25 + 30Q - 9Q^2$
$FC = 55$

Solved ~~at~~ in Past Paper (Fall 2022)

(v)
Given Demand Functions

$$P_d = 25 - Q^2$$

or Supply functions

$$P_s = 2Q + 1$$

Find Consumer or Producer Surplus

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$$P_d = P_s$$

$$25 - Q^2 = 2Q + 1$$

$$0 = Q^3 + 2Q - 25 + 1$$

$$Q^2 + 2Q - 24 = 0$$

$$Q^2 + 6Q - 4Q - 24 = 0$$

$$Q(Q + 6) - 4(Q + 6) = 0$$

$$(Q - 4)(Q + 6) = 0$$

$$\boxed{Q = 4} \qquad \boxed{Q = -6}$$

So $Q_0 = 4$

For P_0 put in P_s

$$P_s = 2Q + 1$$

$$P_0 = 2(4) + 1$$

$$P_0 = 8 + 1$$

$$\boxed{P_0 = 9}$$

Now Consumer Surplus is

$$C.S = Q_0P_0 + \int_0^{Q_0} f(Q)\, dQ - Q_0P_0$$

$$C.S = 4(9) - \int^{4} (25 - Q^2)\, dQ$$

$$C.S = - \int_0^4 25\, dQ - \int_0^4 Q^2\, dQ$$

$$C.S = \left[25Q - \frac{Q^3}{3} \right]_0^4 \quad 36$$

$$C.S = \left[25(4) - 0 - \frac{(4)^3}{3} - 0 \right]$$

$$C.S = -\left[100 - 21.3 \right]$$

$$C.S = 78.6 - 36$$

$$\boxed{C.S = 42.6}$$

Now Producer Surplus

$$P.S = Q_0 P_0 - \int_0^{Q_0} f(Q)\, dQ$$

$$P.S = 36 - 2\int_0^4 Q + 1\, dQ$$

$$P.S = 36 - \left[\frac{\cancel{2} Q^2}{\cancel{2}} + Q \right]_0^4$$

$$P.S = 36 - \left[Q^2 + Q \right]_0^4$$

$$P.S = 36 - \left[(4)^2 - 0 + 4 - 0 \right]$$

$$P.S = 36 - 20$$

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$P.S = 16$

Consumer Surplus = 42.6

Producer Surplus = 16

(vi) Solve differential Equations

$$\frac{dy}{dt} - 6y = 18$$

Solved ~~at~~ in Past Paper (2022 Fall)

Q#2:- LONG QUESTIONS.

i) Solve the Second order linear Equations differential.

$$y''(t) + y'(t) + \frac{1}{4} y_t = 9$$

Subject to $y_0 = 30$, $y'_0 = 15$

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$$Y_p = \frac{b}{a} = \frac{9}{1/4} = 9(4) = 36$$

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$$Y''(t) + Y'(t) + \frac{1}{4} Y(t) = 36$$

$\boxed{\text{We use discrit real root}}$

Case II if $a_1^2 = 4a_2$

$$(1)^2 = 4\left(\frac{1}{4}\right)$$

$$1 = 1$$

So

Genral Solutions.

$$Y_t = Y_c + Y_p$$

For Y_c

we use discrit the (DRRC) II

$$\boxed{Y_c = A_1 e^{rt} + A_2 t e^{rt}}$$

$$Y_c = A_1 e^{rt} + A_2 t e^{rt}$$

For r

$$r^2 + r + c = 0$$

$$r^2 + r + \frac{1}{4} = 0$$

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$$r^2 + r + \frac{1}{4} = 0$$

We Multiply by (4) both side

$$4r^2 + 4r + 4\left(\frac{1}{4}\right) = 4(0)$$

$$4r^2 + 4r + 1 = 0$$

Factraizations:-

$$4r^2 + 2r + 2r + 1 = 0$$

$$2r(2r + 1) + 1(2r+1) = 0$$

$$(2r+1)(2r+1) = 0$$

$$2r + 1 = 0$$

$$\boxed{r = -1/2}$$

Put value of r in Y_c

$$Y_c = A_1 e^{-1/2 t} + A_2 t e^{-1/2 t}$$

$$Y_c = A_1 e^{-t/2} + A_2 t e^{-t/2}$$

If $t = 0$

$$Y_0 = A_1 (e^{-0/2}) + A_2 (0)(e^{-0/2})$$

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$Y_0 = A_1 + 0 \quad \because Y_0 = 30$

$\boxed{30 = A_1}$

For A_2

$Y'(0) = -\frac{1}{2} A_1 e^{-1/2(t)} + \left[A_2 t\left(-\frac{t}{2} e^{-1/2 t}\right) + e^{-1/2 t}(A_2)\right]$

$Y'(c) = -\frac{A_1}{2} e^{-1/2 t} + \quad ''$

If $t = 0$

$Y'(0) = -\frac{A_1}{2}(e^0) + (0 + e^0 A_2)$ $(Y_0' = 15)$

$15 = -\frac{A_1}{2} + A_2$

$15 = -\frac{30}{2} + A_2$

$15 + \frac{30}{2} = A_2$

$A_2 = \frac{30 + 30}{2}$

$\boxed{A_2 = 30}$

Now put A_1 or A_2 value in Y_c (C_1, S)

$Y_c = 30 e^{-1/2 t} + t 30 e^{-1/2 t}$

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$$Y_t = Y_c + Y_p$$

Hence $Y_p = 96$

$$\boxed{Y_t = 30e^{-1/2t} + 30te^{-1/2t} + 36}$$

(iii)

Evaluate the Integral

$$\int x^3 e^x \, dx$$

$$= x^3 \int e^x \, dx - \int \left\{ \frac{d}{dx}(x^3) \int e^x \right\} dx$$

$$= x^3 e^x - \int (3x^2 \cdot e^x) \, dx$$

$$= x^3 e^x - 3\int x^2 \cdot e^x \, dx$$

$$= x^3 e^x - 3\left[x^2 \int e^x dx - \int \left\{ \frac{d}{dx} x^2 \int e^x \right\} dx \right]$$

$$= x^3 e^x - 3\left[x^2 e^x - \int (2x\, e^x) \, dx \right]$$

$$= x^3 e^x - 3\left[x^2 e^x - 2\int (xe^x) \, dx \right]$$

$$= x^3 e^x - 3\left[x^2 e^x - 2\left\{ x \int e^x dx - \int \left(\frac{d}{dx}(x) \int e^x \right) dx \right\} \right]$$

$= x^3e^x - 3\left[x^2e^x - 2\left(xe^x - \int e^x\right) dx\right]$

$= x^3e^x - 3\left[x^2e^x - 2\left(xe^x - e^x\right)\right] + c$

$= x^3e^x - 3x^2e^x + 6xe^x - 6e^x + c$

$= 3xe^x\left(x^2 - x + \cancel{2} - \cancel{2}\right) + c$

$= 3xe^x\left(x^2 - x\right) + c$